

W4 - 1.4 Working with Radicals

MCR3U

1) Simplify

a) $3(4\sqrt{5})$

b) $\sqrt{5}(-2\sqrt{7})$

c) $2\sqrt{3}(3\sqrt{2})$

2) Express each as a mixed radical in simplest form

a) $\sqrt{12}$

b) $\sqrt{147}$

c) $\sqrt{252}$

3) Simplify

a) $2\sqrt{3} - 5\sqrt{3} + 4\sqrt{3}$

b) $11\sqrt{5} - 4\sqrt{5} - 5\sqrt{5} - 6\sqrt{5}$

c) $\sqrt{6} - 4\sqrt{2} + 3\sqrt{6} - \sqrt{2}$

d) $2\sqrt{10} - \sqrt{10} - 4\sqrt{10} + \sqrt{5}$

4) Add or subtract as indicated

a) $8\sqrt{2} - 4\sqrt{8} + \sqrt{32}$

b) $\sqrt{20} - 4\sqrt{12} - \sqrt{125} + 2\sqrt{3}$

c) $5\sqrt{3} - \sqrt{72} + \sqrt{243} + \sqrt{8}$

d) $\sqrt{44} + \sqrt{88} + \sqrt{99} + \sqrt{198}$

5) Expand and simplify

a) $5\sqrt{6}(2\sqrt{3})$

b) $8\sqrt{5}(\sqrt{10})$

c) $11\sqrt{2}(5\sqrt{3})$

6) Expand and simplify where possible

a) $3(8 - \sqrt{5})$

b) $\sqrt{3}(\sqrt{6} - \sqrt{3})$

c) $8\sqrt{2}(2\sqrt{8} + 3\sqrt{12})$

7) Expand and simplify where possible

a) $(\sqrt{2} + 5)(\sqrt{2} + 5)$

b) $(\sqrt{3} + 2\sqrt{2})(5 + 5\sqrt{2})$

c) $(1 + \sqrt{5})(1 - \sqrt{5})$

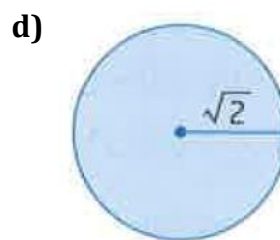
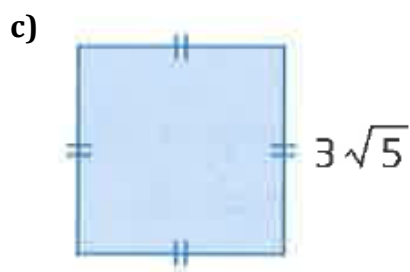
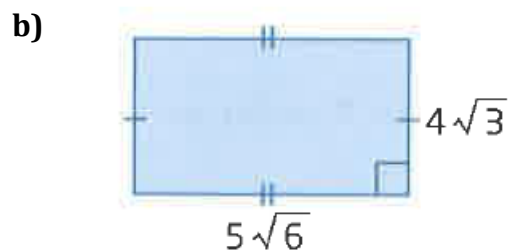
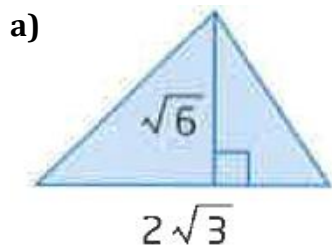
d) $(4 - 3\sqrt{7})(\sqrt{7} + 1)$

8) Simplify

a) $\frac{1}{4}\sqrt{54} - \frac{1}{4}\sqrt{150}$

b) $\frac{1}{2}\sqrt{8} + \frac{3}{5}\sqrt{50} - \frac{2}{3}\sqrt{18}$

9) Find a simplified expression for the area of each shape



10) Find the area and perimeter of the rectangle shown. Express your answer in simplified radical form.



11) Simplify each of the following

a) $\frac{21-7\sqrt{6}}{7}$

b) $\frac{12-\sqrt{48}}{4}$